

AR Lecture 1



Introduction to Immersive Technologies

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What is Reality?

- Human perception of the **physical world** through senses.
- Limitations of natural perception.



Why AR/VR/MR (XR)?

- Extend, enhance, or replace real-world perception.
- Enable **simulation, training, visualization, and interaction.**



XR Phase 1 - VR

- A fully computer-generated environment that replaces the real world and allows user interaction.



XR Phase 2 - AR

- Overlays digital information (text, images, 3D objects) onto the real-world environment.



XR Phase 3 - MR

- Combines real and virtual worlds where digital objects interact with the real environment in real time.



Evolution of XR

- **Early Concepts:**
 - 1838 – Stereoscope (Charles Wheatstone)
 - 1950s – Sensorama (Morton Heilig)
- **Development Phase:**
 - 1968 – First Head-Mounted Display (Ivan Sutherland)
 - 1990s – VR used in military and research labs
- **Modern Era:**
 - 2012 – Oculus Rift revival
 - 2016 – Pokémon Go popularizes AR
 - 2020+ – Industrial, medical, and educational MR systems

Reality–Virtuality Continuum

- **Concept:**
 - Proposed by **Milgram & Kishino (1994)**
- **Continuum Levels:**
 - Real Environment
 - Augmented Reality (AR)
 - Mixed Reality (MR)
 - Augmented Virtuality
 - Virtual Reality (VR)
- **Explanation:**
 - As we move from left to right, **virtual content increases**.
 - MR lies between AR and VR.



Milestones in AR/VR History

- **Key Milestones:**
- **1968:** Sword of Damocles (first HMD)
- **1994:** Reality–Virtuality Continuum defined
- **2012:** Oculus Rift Kickstarter
- **2016:** Pokémon Go
- **2019:** Microsoft HoloLens 2
- **2023+:** Metaverse & AI-powered XR

Comparisons

Feature	AR	VR	MR
Real World Visible	Yes	No	Yes
Immersion Level	Low	High	Medium–High
Devices	Mobile, Glasses	Headsets	HoloLens
Interaction	Limited	Full	Advanced

Recap

- **Quick Recap:**
 - AR adds to reality
 - VR replaces reality
 - MR blends both
 - Continuum explains transition